



Cisco Meeting Server (CMS)

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Agenda

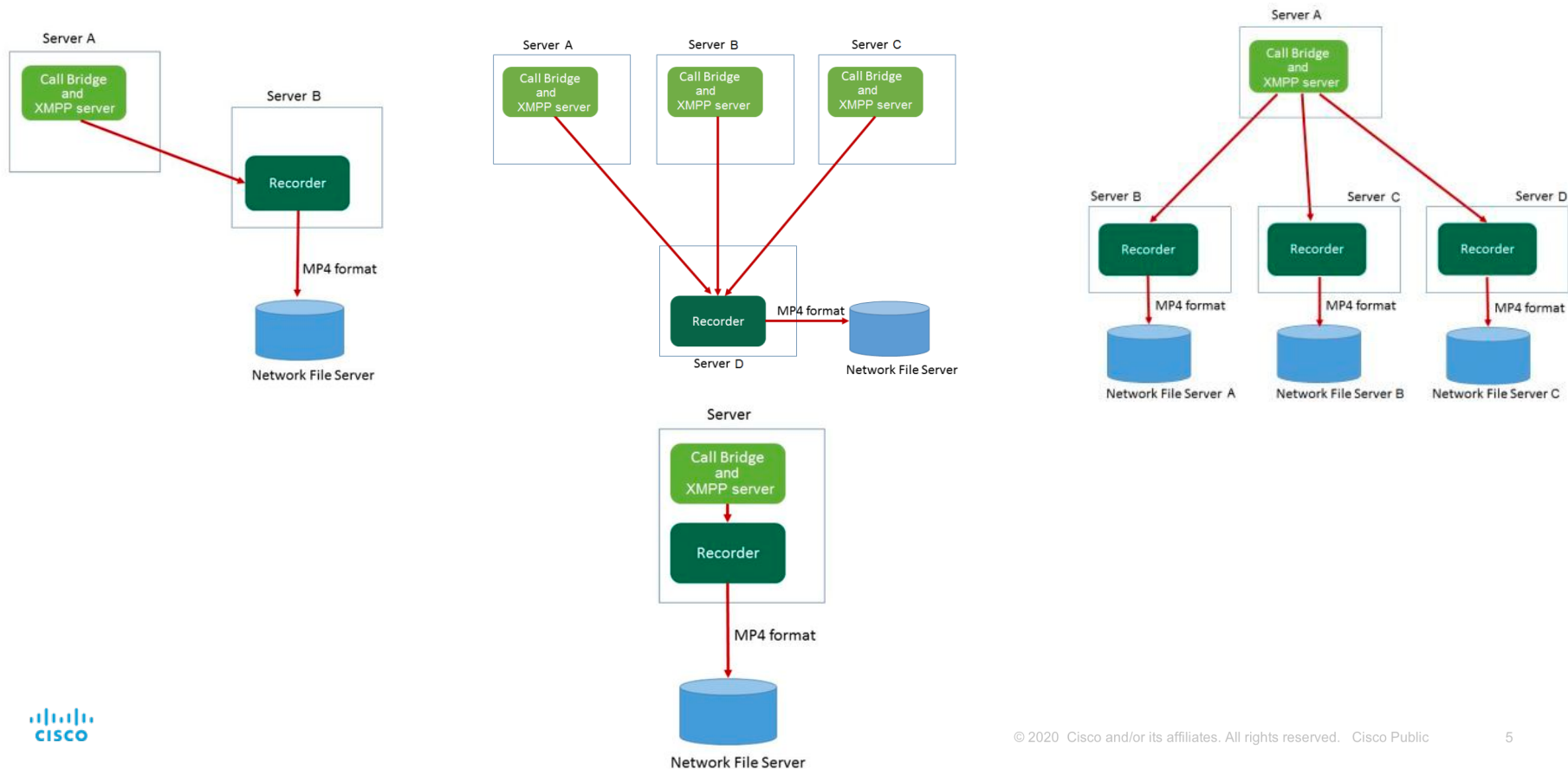
- Recording
- Streaming
- Cisco Meeting Management (CMM)

Recording

What is a recorder?

- CMS Recorder enables the capability to record meetings on CoSpaces.
- The transcoding takes place real-time on the Recorder Server, while writing a temp file to the NFS
- At the end of the call, the file is automatically converted into an MP4 format and stored on NFS.

Supported Deployments



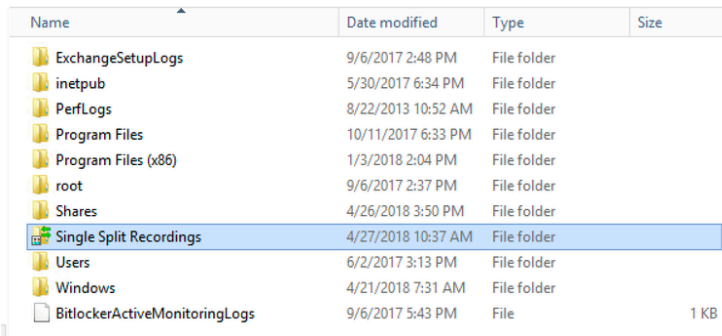
Recorder Server Requirements:

- Recommended to run recorder on dedicated server.
- 4 Physical Cores and 4GB RAM
- 1 Core will support 2 Simultaneous recordings, using 0.5 GB of Memory per recording.
- Can support 8 simultaneous recordings.

Component	Purpose	vCPU	RAM	Minimum CPU	Maximum CPU
Recorder	2 recordings	1	1 GB (0.5 GB per)	4	24

Recording

Step 1. Configure an NFS Share Folder on a Windows Server



Name	Date modified	Type	Size
ExchangeSetupLogs	9/6/2017 2:48 PM	File folder	
inetpub	5/30/2017 6:34 PM	File folder	
PerfLogs	8/22/2013 10:52 AM	File folder	
Program Files	10/11/2017 6:33 PM	File folder	
Program Files (x86)	1/3/2018 2:04 PM	File folder	
root	9/6/2017 2:37 PM	File folder	
Shares	4/26/2018 3:50 PM	File folder	
Single Split Recordings	4/27/2018 10:37 AM	File folder	
Users	6/2/2017 3:13 PM	File folder	
Windows	4/21/2018 7:31 AM	File folder	
BitlockerActiveMonitoringLogs	9/6/2017 5:43 PM	File	1 KB

Step 2. Configure and enable recorder on the Recorder server

1. Configure the Recorder to listen on the interface:

recorder listen a:8443

If the recorder is on the local CB, the interface must be the set to “loopback”:

recorder listen lo:8443

2. Set the certificate file to be used by the recorder:

recorder certs <key file> <certificate file>

3. Add the Call bridge certificates bundle to the Recorder trust store:

recorder trust <crt-bundle>

4. Specify the hostname/IP address of the NFS, and the directory on the NFS to Store the recordings:

recorder nfs <hostname/IP>:<directory>

5. Now enable the Recorder:

recorder enable

```
acanosrv01> recorder
Enabled                : true
Interface whitelist    : lo:8443
Key file               : callbridgecert.key
Certificate file       : callbridgecert.cer
Trust bundle           : callbridgecert.cer
NFS domain name       : 10.48.36.246
NFS directory          : /acano
```

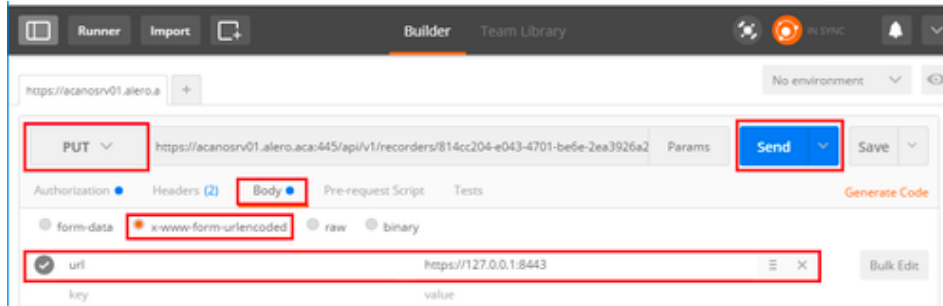

Step 3. Add the Recorder onto the CB using the API

1. Send an **empty POST** with `https://<Callbridge_IP>:445/api/v1/recorders`
2. Then, send a GET with the **same URL** in (a), copy the recorder ID
3. Set the URL of the recorder by sending a PUT with `https://<Callbridge_IP>:445/api/v1/recorders/<recorderid>` and add this in BODY before you execute the PUT:

url=<https://127.0.0.1:8443> (if the recorder is on the local CB)

or

url=`https://<IP Address of recorder>:8443` (if the recorder is not on the local CB)



In case of CMS 2.9 or above,
you can do that from **Configuration > API > /api/vi/recorders**

Status ▾ Configuration ▾ Logs ▾

API objects

This page shows a list of the objects supported by the API. Where you see a ► control, yo

Filter (3 of 116 nodes)

/api/v1/recorders ◀

« start ◀ prev 1 - 1 (of 1) next »

Create new Table view XML view



Status ▾ Configuration ▾ Logs ▾

« return to object list

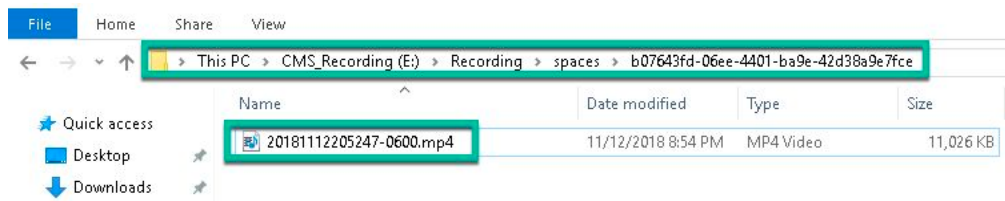
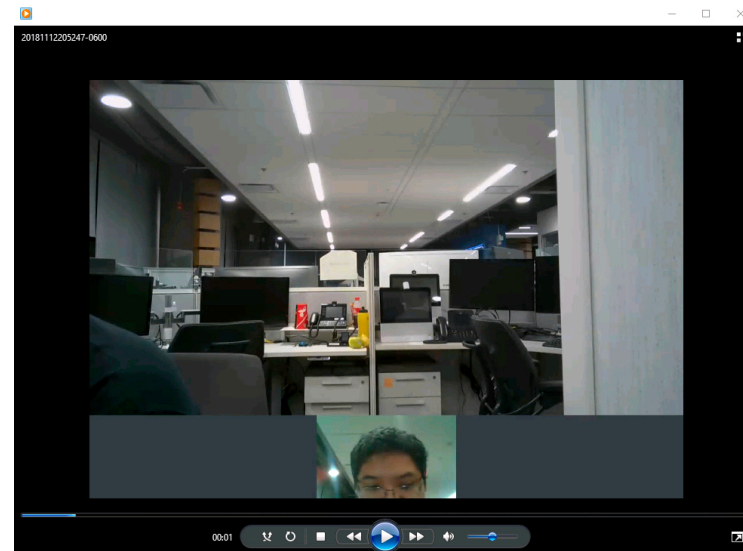
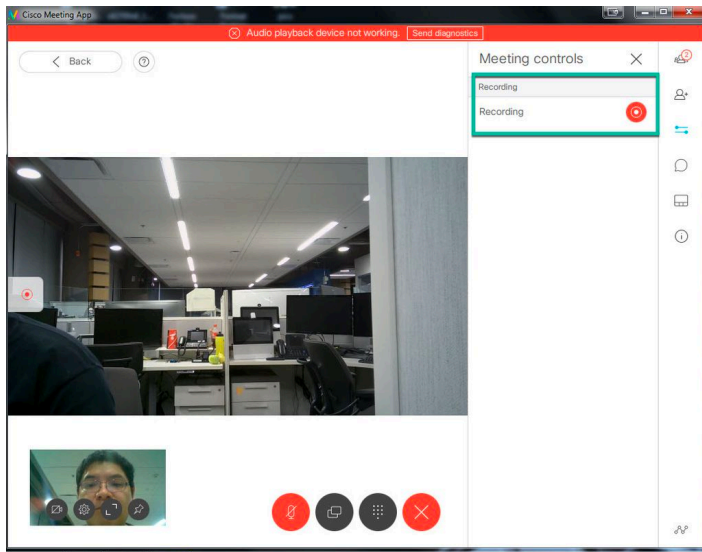
/api/v1/recorders

url	☒	https://127.0.0.1:8443	(URL)
callBridge	☐		Choose
callBridgeGroup	☐		Choose

Create



Step 4. Join meeting using CMA App and use “Record” button to start recording.

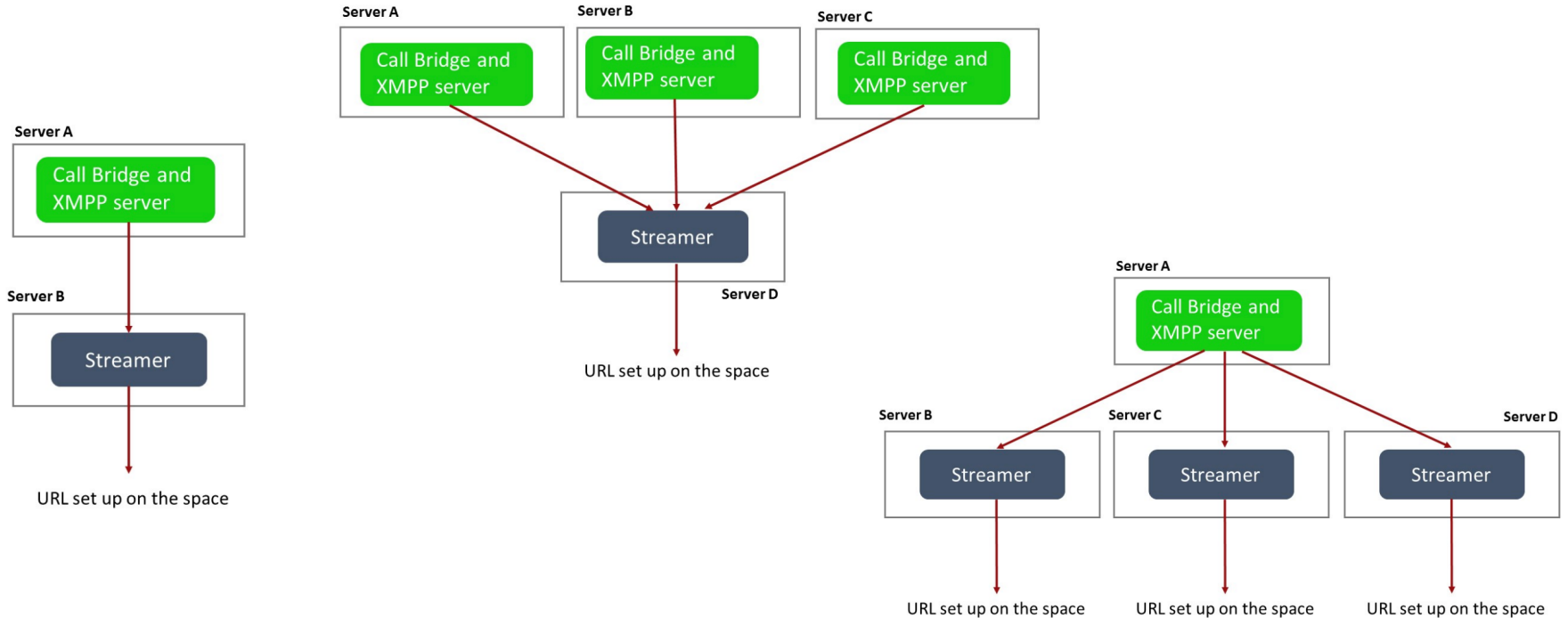


Streaming

What is a Streamer?

- The Streamer enables the capability of meetings in Co-Spaces to be streamed.
- The Streamer sends the media to an external streaming server based on URI configured on CoSpace.
- RTMP is the supported protocol used by the streamer.

Supported Deployments



Streamer Server Requirements:

- Recommended to run on dedicated or combined with recorder
- 4 Physical Cores and 4GB RAM
- 1 Core will support 6 Simultaneous streams.

Component	Purpose	vCPU	RAM	Minimum CPU	Maximum CPU
Streamer	6 concurrent streams	1	1 GB	4	32

Streaming

Step 1. Configure interface(s) for the streamer to listen:

streamer listen a:8443

Step 2. Set the certificate file to be used by the streamer:

streamer certs streamer.key streamer.crt

Step 3. Add the Call bridge certificates bundle to the streamer trust store:

streamer trust callbridge.crt

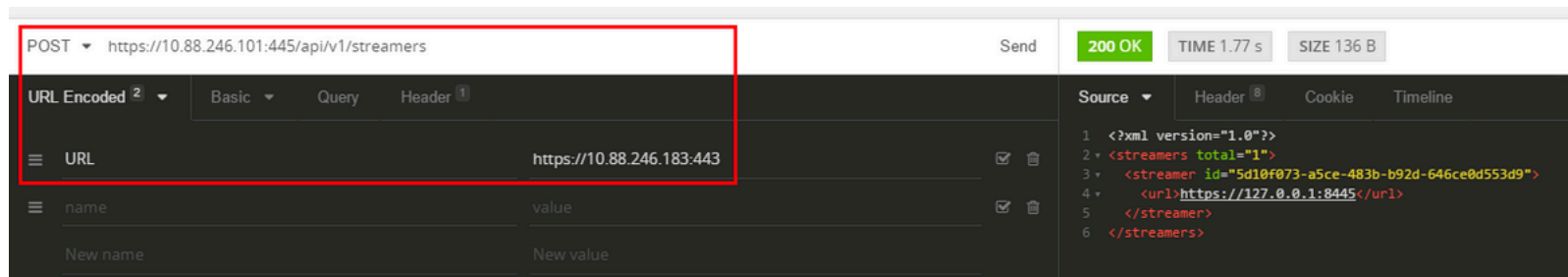
Step 4. Now enable the streamer:

streamer enable

```
CMSCombined> streamer
Enabled                : true
Interface whitelist    : a:8447
Key file               : cmscomb.key
Certificate file       : certnew.cer
CA Bundle file        : RootCA.cer
Trust bundle          : certnew.cer
CMSCombined> █
```


Step 5. Add the streamer onto the CB using the API

1. Send a “POST” to `https://<Callbridge_IP>:445/api/v1/streamers/` with below
`url=https://<IP Address of streamer >:<port>`



The screenshot shows a REST client interface with a POST request to `https://10.88.246.101:445/api/v1/streamers`. The request is successful, returning a 200 OK status with a response time of 1.77 s and a size of 136 B. The response body is XML, indicating that one streamer was added successfully. The streamer's ID is `5d10f073-a5ce-483b-b92d-646ce0d553d9` and its URL is `https://127.0.0.1:8445`.

```
POST https://10.88.246.101:445/api/v1/streamers
200 OK
TIME 1.77 s
SIZE 136 B

URL Encoded 2 Basic Query Header 1
URL https://10.88.246.183:443
name value
New name New value

Source Header 8 Cookie Timeline
1 <?xml version="1.0"?>
2 <streamers total="1">
3 <streamer id="5d10f073-a5ce-483b-b92d-646ce0d553d9">
4 <url>https://127.0.0.1:8445</url>
5 </streamer>
6 </streamers>
```

Step 6. Add the streamer to a CoSpace.

1. Send “GET” `https://<CMS IP>:<port>/api/v1/Cospaces` to get list of CoSpace you have created along with their ID's. Here copy the CoSpace ID to which you want to assign this streamer to.

In case of CMS 2.9 or above,
you can do that from **Configuration > API > /api/vi/streamers**

Status ▼ Configuration ▼ Logs ▼

API objects

This page shows a list of the objects supported by the API. Where you see a ► control, you can enter

Filter (3 of 116 nodes)

/api/v1/streamers ◀

« start ◀ prev 1 - 1 (of 1) next ▶

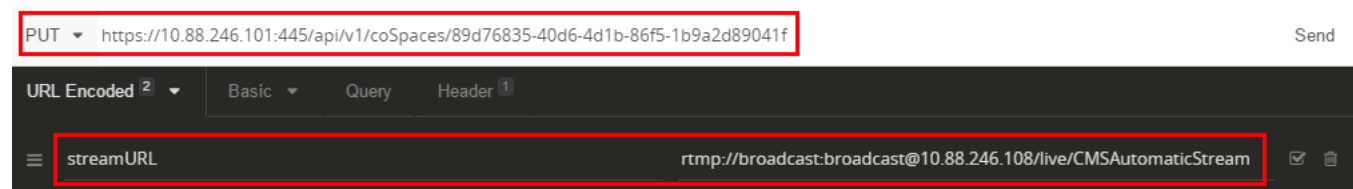
Status ▼ Configuration ▼ Logs ▼

◀ return to object list

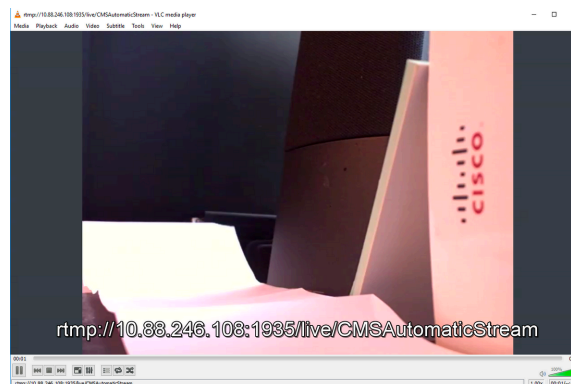
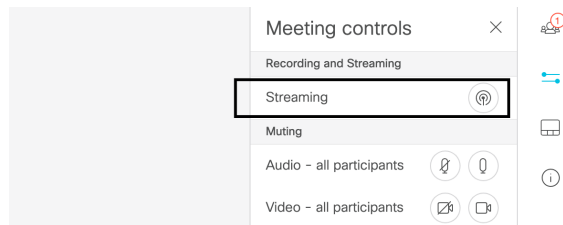
/api/v1/streamers

url	☒	https://127.0.0.1:8445	(URL)
callBridge	☐		Choose
callBridgeGroup	☐		Choose
Create			

2. Sent “PUT” `https://<CMS IP>:<Port>/api/v1/Cospaces/<CoSpace ID>` with below
streamURL = `rtmp://<VBrickBroadcastUsername>:<VBrickBroadcastPassword>@<VBrick IP or FQDN>/live/NameoftheStream`



Step 7: Join the meeting from CMA, Click on “Streaming” to start streaming and verify stream on using stream URL.



Cisco Meeting Management

What is CMM?

- Cisco Meeting Management is a management tool for Cisco's on-premises video meeting platform, Cisco Meeting Server.
- You can use the tool to monitor and manage meetings that are running on the platform, and it also provides information about which Cisco licenses you are using.

VM-Requirements

- Meeting Management can manage anything from a single Call Bridge to multiple clustered deployments.
- The VM requirements depend on your deployment size. See the capacity table below to determine your deployment size.

	Small to medium deployments	Large deployments
Call Bridges	1-8 Call Bridges on Cisco Meeting Server 1000 or 1 Call Bridge on Cisco Meeting Server 2000 or 1-4 Call Bridges run on X3 server	9-24 Call Bridges run on Cisco Meeting Server 1000 or 2-3 Call Bridges run on Cisco Meeting Server 2000 or 5-10 Call Bridges run on X3 server
Call legs started (at peak time, across all Call Bridges)	10 call legs started per second	20 call legs started per second
Users signed in to Meeting Management at the same time	15 concurrent users	25 concurrent users
Meetings per week (across all Call Bridges)	10,000	10,000

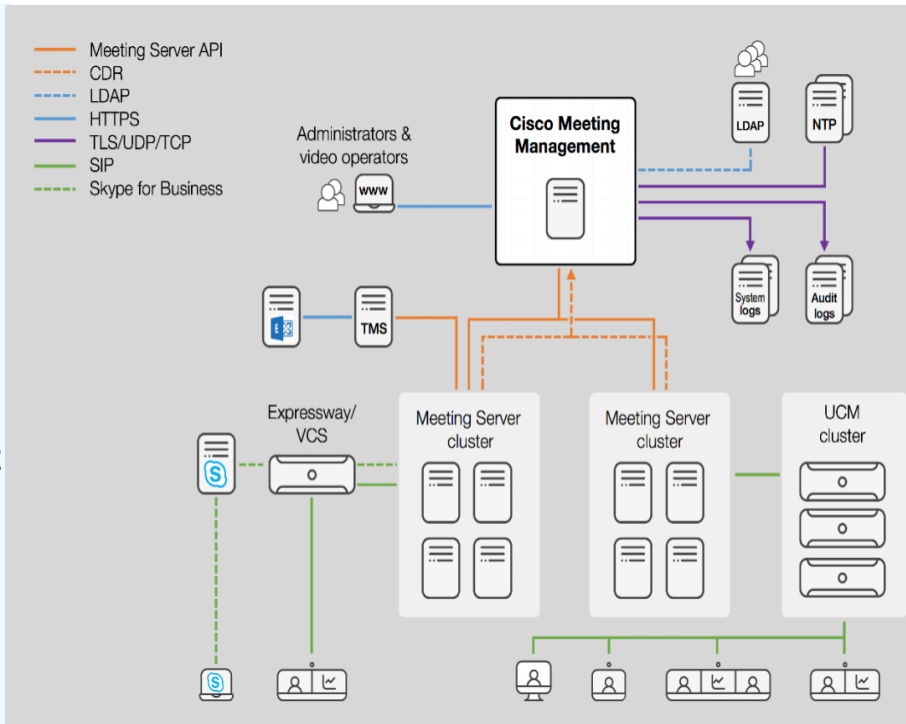
VM-Requirements

- The VM is configured for small to medium deployments. For large deployments, you must change the sizing manually during setup.
- If you have a medium size deployment and think you may need higher capacity later, then configure your VM for a large deployment

Requirement	Small and medium deployments	Large deployments
Server manufacturer	Any	Any
Processor type	Intel / AMD	Intel / AMD
Processor frequency	2.0 GHz	2.0 GHz
vCPU	4 cores	8 cores
Storage	100 GB <i>We recommend thick provisioning and eager zeroing.</i>	100 GB <i>We recommend thick provisioning and eager zeroing.</i>
RAM	4 GB reserved memory	8 GB reserved memory
Hypervisor	ESXi 6.0 U3, ESXi 6.5, ESXi 6.7	ESXi 6.0 U3, ESXi 6.5, ESXi 6.7
Network interfaces	1	1

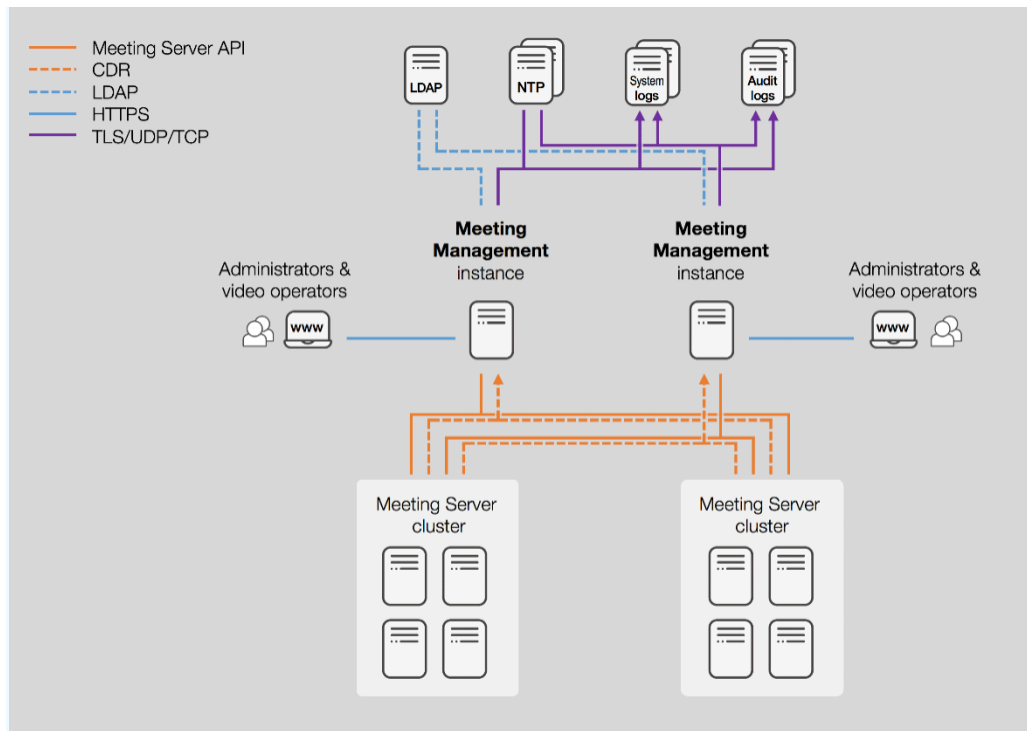
CMM-Single Deployment

- Single instance of CMM can manage deployments from single CallBridge to Cluster of CallBridge
- NTP servers synchronize Meeting Servers and Meeting Management instances
- External syslog server for system and audit logs
- User access is authenticated via LDAP



CMM- Resiliency Deployment

- Up to 2 instances of CMM supported for added resiliency
- CMM instances configured independently (active: active)
- Recommended to deploy the 2 instances in different locations
- No information is exchanged between the two CMM
- Create a certificate for each instance of CMM
- Each certificate must include both the user facing address and the unique CDR receiver address



Certificate Requirement-CMM

- Meeting Management uses a certificate **to identify itself to browsers and to Call Bridges**.
- Setup, Meeting Management generates a self-signed certificate which you can use during initial configuration.
- In a production environment, you must **replace the self-signed certificate with a certificate signed by a CA** (Certificate Authority). You can use an internal or external CA, depending on the requirements in your organization.
- The certificate chain **should include** the certificate of the **CA** that signed the certificate, plus any certificates higher in the certificate chain, up to and including the root CA certificate.
- Your **CDR receiver address**, as well as any addresses your users will use for the browser interface, should be included in the certificate.
- When the SAN field is used, Meeting Management does not look at the Common Name. The CDR receiver address must be included in the SAN field.
- Meeting Management has no capability to create certificate signing requests. **Use a dedicated tool, for instance the OpenSSL toolkit, to create your private key and a certificate signing request.** CMS server can also be used.

CMM-LDAP & Users

- Meeting Management supports locally managed users as well as user authentication via LDAP
- Local users are added and managed locally on the Meeting Management Users page. These users are authenticated directly by Meeting Management.
- LDAP users are added via mappings to existing groups on your LDAP server. Meeting Management uses your LDAP server to authenticate these users by checking their group membership when they sign in
- Authentication via LDAP is recommended for general use and administration

Users can have two roles:

Administrators have full access to Meeting Management. Administrators will typically set up Meeting Management, change configurations, add users, and monitor and maintain the system.

Video operators only have access to the Meetings and Overview pages. Video operators monitor and manage meetings, and they perform basic troubleshooting related to ongoing meetings. For instance, they may try to call a participant who got disconnected or check the call statistics if someone has audio issues.

CMM-Feature Overview

- See list of active or recent meetings
- Search by meeting title or individual participant
- Pin meeting at top of the list
- See connected Cisco Meeting Servers
- Add and remove Call Bridge nodes or clusters

The screenshot displays the Cisco Meeting Management (CMM) web interface. The top navigation bar includes the Cisco logo, 'Cisco Meeting Management', a notifications bell icon, and the user 'Local/Seeta Rama Administrator'. A left sidebar contains icons for Overview, Meetings, Users, and Servers.

The 'Meetings' section is active, showing a search bar and filters for Type (Scheduled, Unscheduled), Status (Active, Upcoming, Ended), and a 'Clear filters' button. It indicates 'Showing 1 of 15 meetings'. A table lists meeting details:

Meeting title	Owner	Participants	Activity	Start time	End time	Duration
Test_Slot_3		2		07/07/2020 11:47		00:02:11

The 'Servers' section shows a cluster '<CMS_cluster>' associated with TMS, with links for 'Set up provisioning', 'Associate cluster with TMS', 'Rename cluster', and 'Remove cluster'. Below is a table of discovered servers:

Display name	Server address	Port	HTTPS	Version	Actions
✓ CMS TMS	cms.anandube.com	445	Verified	2.9.2	

A note indicates '1 auto-discovered Call Bridge - show'. A secondary cluster '<CMS Secondary_cluster>' is also shown with similar links and a table of servers:

Display name	Server address	Port	HTTPS	Version	Actions
✓ CMS Secondary TMS	cmssec.anandube.com	445	Verified	2.9.2	

CMM-Feature Overview - Participant Detail

- View connection details per participant
- Mute audio/video individually
- Drop active participant from meeting
- Per participant layout change/override conference wide setting
- Detailed audio and video call information.
- Move participants in CMM from one space to other.

< ❌ ✔️ BlackBelt

Owner
Start time 07/05/2020 14:09
End time
Duration 00:01:30
Lobby Empty

Activity Important Speaking Status Connected Moved Disconnected

Showing 2 of 2 participants

	1	Participant name	Join time	Leave time	Actions
☒	❌	1234@anandube.com	07/05/2020 14:09		☐ ☐ ☐ ☐ ☐
☐	❌	1235@anandube.com	07/05/2020 14:09		☐ ☐ ☐ ☐ ☐

< ❌ ✔️ 9981896

Owner
Start time 06/16/2020 08:53
End time
Duration 00:32:11
Lobby Empty

Activity Important Speaking Status Connected Moved Disconnected

Showing 1 of 1 participants

	1	Participant name
☒	❌	10.104.215.196

Move Participants

Moving from: 9981896

Participants Selected All

Search participants

Deselect all

10.104.215.196

Moving to

Search participants

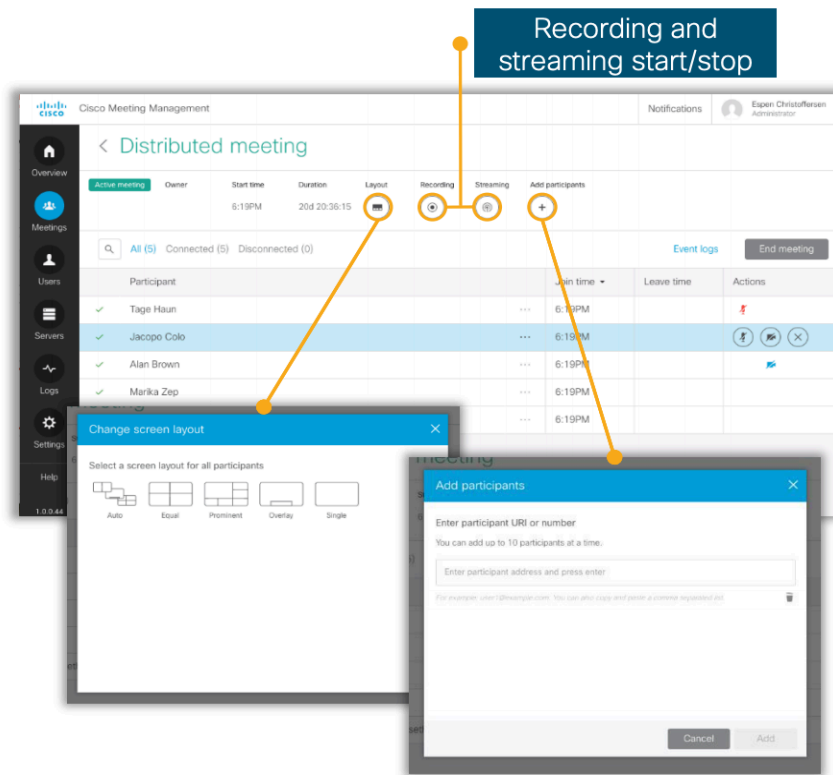
9991 Start time: 06/16/2020 08:49

Moving 10.104.215.196 from 9981896 to 9991

Cancel Move

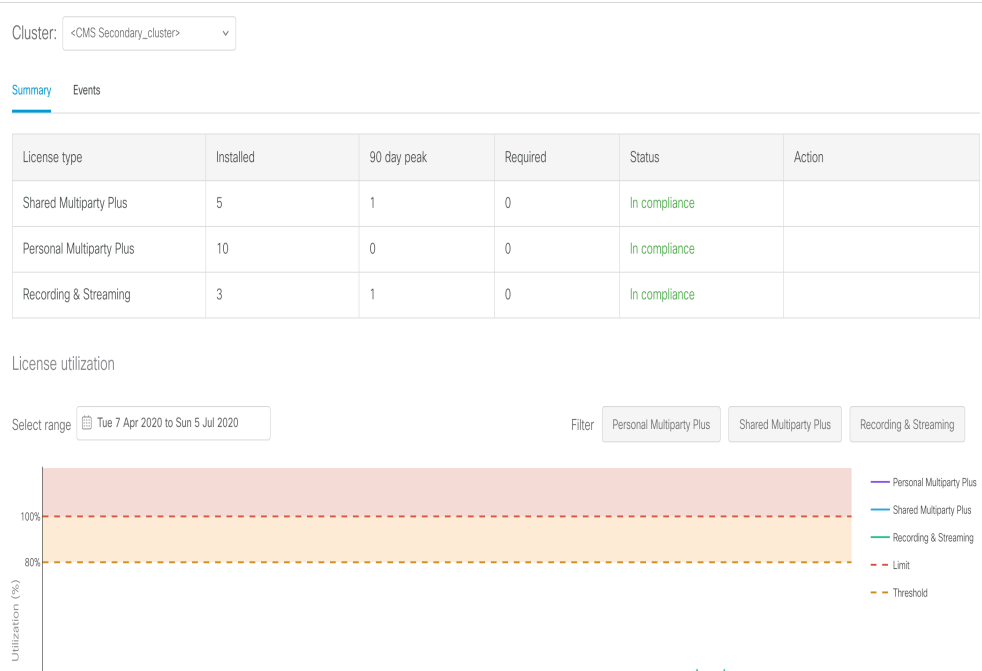
CMM-Feature Overview - Meeting Detail

- List overview of all participants in a meeting
- Conference wide layout change per meeting
- Drop active participant from meeting
- Start and stop recording and streaming *
- Controls to mute audio/video and drop participants
- View and download event log for a meeting



CMM-CMS Licenses

- List all user licenses Including recorder and streamer
- List licenses utilization
- Shows graphical utilization of licenses
- List the licenses events



Questions?